SDOM Standard Inputs - CopperPlate Mode

This document describes every CSV input expected by SDOM when running in CopperPlate (single-area) mode. Dummy CSV files matching this schema are provided in the same folder for reference. For full details, see the SDOM User Guide section 'Input Data'.

formulations.csv

Selects the modeling approach for each major system component.

Column	Type	Description
`Component`	string	Component name (Thermal, Hydro, Imports, Exports, Network).
`Formulation`	string	Formulation assigned to that component (see user guide).
`Description`	string	Optional free-text description.

Load_hourly.csv

System hourly electricity demand time-series.

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`Load`	float	System electricity demand at each hour (MWh).

CFSolar.csv

Hourly solar capacity factors per candidate site.

Column	Type	Description
`Hour`	int	Hour index of the year (1..8760).
	float	Capacity factor [0, 1] for each candidate solar site.

CapSolar.csv

Candidate solar PV sites with capacity bounds and costs.

Column	Type	Description
`sc_gid`	string	Unique identifier for the candidate site.
`capacity`	float	Upper bound for installed capacity at the site (MW).
`latitude`	float	Latitude of the site (optional).
`longitude`	float	Longitude of the site (optional).
`trans_cap_cost`	float	Transmission interconnection capex (USD/kW).
`CAPEX_M`	float	Capital expenditure (USD/kW).
`FOM_M`	float	Fixed O&M; cost (USD/kW-yr).

CFWind.csv

Hourly wind capacity factors per candidate site.

Column	Type	Description
`Hour`	int	Hour index of the year (1..8760).
	float	Capacity factor [0, 1] for each candidate wind site.

CapWind.csv

Candidate wind sites with capacity bounds and costs (same schema as CapSolar.csv).

Column	Type	Description
`sc_gid`	string	Unique identifier for the candidate site.
`capacity`	float	Upper bound for installed capacity at the site (MW).
`latitude`	float	Latitude of the site (optional).
`longitude`	float	Longitude of the site (optional).
`trans_cap_cost`	float	Transmission interconnection capex (USD/kW).
`CAPEX_M`	float	Capital expenditure (USD/kW).
`FOM_M`	float	Fixed O&M; cost (USD/kW-yr).

Nucl_hourly.csv

Fixed nuclear generation profile.

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`Nuclear`	float	Nuclear generation at each hour (MWh).

lahy_hourly.csv

Hydropower hourly time-series (profile or budget basis).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`LargeHydro`	float	Hydro generation or budget basis (MWh).

lahy_max_hourly.csv

Maximum hourly hydro capacity (used by budget formulations).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`LargeHydro`	float	Upper bound on hydro dispatch (MW).

lahy_min_hourly.csv

Minimum hourly hydro generation (used by budget formulations).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`LargeHydro`	float	Lower bound on hydro dispatch (MW).

otre_hourly.csv

Other renewable generation profile (geothermal, biomass, etc.).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`OtherRenewables`	float	Other-renewable generation at each hour (MWh).

StorageData.csv

Technical and economic parameters per storage technology (rows = parameters, columns = techs).

Column	Type	Description
`P_Capex`	float	Power-related capital expenditure (USD/kW).
`E_Capex`	float	Energy-related capital expenditure (USD/kWh).
`Eff`	float	Round-trip efficiency (fraction).
`Min_Duration`	float/int	Minimum storage duration (h).
`Max_Duration`	float/int	Maximum storage duration (h).
`Max_P`	float/int	Maximum power capacity (kW).
`MaxCycles`	int	Maximum number of charge/discharge cycles.
`Coupled`	int (0/1)	1 if input and output power are coupled, else 0.
`FOM`	float	Fixed O&M; cost (USD/kW/year).
`VOM`	float	Variable O&M; cost (USD/MWh).
`Lifetime`	int	Expected system lifetime (years).
`CostRatio`	float	Cost allocation between input/output power (0.5 = equal split).

Data_BalancingUnits.csv

Thermal / balancing unit technical and economic parameters.

Column	Type	Description
`Plant_id`	string	Unique plant or aggregation identifier.
`MinCapacity`	float	Minimum installed capacity (MW).
`MaxCapacity`	float	Maximum installed capacity (MW).
`Lifetime`	int	Operational lifetime (years).
`Capex`	float	Capital expenditure (USD/kW).

Column	Type	Description
`HeatRate`	float	Heat rate (MMBtu/MWh).
`FuelCost`	float	Fuel cost (USD/MMBtu).
`VOM`	float	Variable O&M; cost (USD/MWh).
`FOM`	float	Fixed O&M; cost (USD/kW).

scalars.csv

Global scalar parameters for the optimization.

Column	Type	Description
`LifeTimeVRE`	int	Operational lifetime of VRE assets (years), used in CRF.
`GenMix_Target`	float	Target renewable generation share [0, 1].
`AlphaNuclear`	int (0/1)	Activate (1) or deactivate (0) nuclear.
`AlphaLargHy`	int (0/1)	Activate (1) or deactivate (0) large hydro.
`AlphaOtheRe`	int (0/1)	Activate (1) or deactivate (0) other renewables.
`r`	float	Discount / interest rate (e.g., 0.06).
`EUE_max`	float	Maximum allowed Expected Unserved Energy (resiliency).

Import_Cap.csv

Hourly import capacity limits (required with CapacityPriceNetLoadFormulation for Imports).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`Import`	float	Import capacity at each hour (MW).

Import_Prices.csv

Hourly import prices (required with CapacityPriceNetLoadFormulation for Imports).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`Import`	float	Import price at each hour (USD/MWh).

Export_Cap.csv

Hourly export capacity limits (required with CapacityPriceNetLoadFormulation for Exports).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`Export`	float	Export capacity at each hour (MW).

Export_Prices.csv

Hourly export prices (required with CapacityPriceNetLoadFormulation for Exports).

Column	Type	Description
`*Hour`	int	Hour index of the year (1..8760).
`Export`	float	Export price at each hour (USD/MWh).